

Boss Challenge — Paper 13 (Class 7)

Respiration in Organisms | Transportation in Animals & Plants | Fractions & Decimals | Perimeter & Area
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The process by which cells break down food to release energy is called:
(A) photosynthesis
(B) transpiration
(C) digestion
(D) cellular respiration
2. The fraction $\frac{6}{8}$ written in its lowest terms is:
(A) $\frac{2}{4}$
(B) $\frac{6}{8}$
(C) $\frac{3}{4}$
(D) $\frac{4}{3}$
3. The liquid that carries food, oxygen and wastes around the human body is:
(A) saliva
(B) bile
(C) blood
(D) urine
4. The perimeter of a square of side 6 cm is:
(A) 30 cm
(B) 36 cm
(C) 24 cm
(D) 12 cm
5. The gas that living cells take in during normal breathing to release energy is:
(A) hydrogen
(B) oxygen
(C) nitrogen
(D) carbon dioxide
6. The decimal 0.25 written as a fraction in lowest terms is:
(A) $\frac{2}{5}$
(B) $\frac{1}{2}$
(C) $\frac{1}{4}$
(D) $\frac{25}{10}$
7. The muscular organ that pumps blood around the body is the:
(A) kidney
(B) liver
(C) heart
(D) lungs

8. The area of a square of side 6 cm is:
- (A) 24 cm²
 - (B) 18 cm²
 - (C) 12 cm²
 - (D) 36 cm²
9. Breakdown of glucose without oxygen, as in our muscles during heavy exercise, is called:
- (A) complete combustion
 - (B) fermentation in plants
 - (C) aerobic respiration
 - (D) anaerobic respiration
10. Which of these fractions is the largest?
- (A) $\frac{2}{3}$
 - (B) $\frac{3}{5}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{3}{4}$
11. The red colour of blood and its ability to carry oxygen are due to:
- (A) plasma
 - (B) white cells
 - (C) haemoglobin
 - (D) platelets
12. The area of a rectangle 8 cm long and 5 cm wide is:
- (A) 26 cm²
 - (B) 40 cm²
 - (C) 13 cm²
 - (D) 80 cm²
13. The partly-broken-down substance that builds up in muscles and causes cramps is:
- (A) carbon dioxide
 - (B) lactic acid
 - (C) glucose
 - (D) alcohol
14. The sum $\frac{1}{4} + \frac{1}{2}$ equals:
- (A) $\frac{2}{4}$
 - (B) $\frac{3}{4}$
 - (C) $\frac{2}{6}$
 - (D) $\frac{1}{6}$
15. Blood vessels that carry blood away from the heart to the body are called:
- (A) arteries
 - (B) capillaries
 - (C) veins
 - (D) nerves

16. The perimeter of a rectangle 8 cm long and 5 cm wide is:
- (A) 26 cm
 - (B) 52 cm
 - (C) 13 cm
 - (D) 40 cm
17. In yeast, anaerobic respiration produces carbon dioxide and:
- (A) water only
 - (B) oxygen
 - (C) alcohol
 - (D) lactic acid
18. The product $\frac{2}{3} \times \frac{3}{4}$ equals:
- (A) $\frac{5}{7}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{6}{7}$
 - (D) $\frac{2}{4}$
19. Thin-walled vessels that carry blood back towards the heart and have valves are:
- (A) alveoli
 - (B) capillaries
 - (C) arteries
 - (D) veins
20. A triangular scarf has a base of 14 cm and a height of 9 cm. Its area works out to:
- (A) 63 cm
 - (B) 63 cm^2
 - (C) 23 cm^2
 - (D) 126 cm^2
21. The opening of the windpipe through which air enters from the throat begins at the:
- (A) larynx
 - (B) stomach
 - (C) kidney
 - (D) diaphragm
22. Dividing by a fraction is the same as multiplying by its:
- (A) double
 - (B) opposite
 - (C) square
 - (D) reciprocal
23. The tiny, thin vessels where oxygen and food pass from blood into cells are:
- (A) arteries
 - (B) tendons
 - (C) veins
 - (D) capillaries

24. The area of a parallelogram with base 12 cm and height 5 cm is:

- (A) 17 cm²
- (B) 34 cm²
- (C) 60 cm²
- (D) 30 cm²

25. The muscular sheet below the lungs that helps in breathing by moving up and down is the:

- (A) heart
- (B) liver
- (C) diaphragm
- (D) ribs only

26. The value of $1/2 \div 1/4$ is:

- (A) $1/2$
- (B) 2
- (C) $1/8$
- (D) 4

27. The number of times the heart beats in a minute is measured as the:

- (A) blood pressure
- (B) blood group
- (C) breathing rate
- (D) pulse rate

28. A tile maker needs to know how many square centimetres make up one whole square metre.
The answer is:

- (A) 100 cm²
- (B) 1000000 cm²
- (C) 1000 cm²
- (D) 10000 cm²

29. Tiny balloon-like air sacs in the lungs where gases are exchanged with blood are called:

- (A) alveoli
- (B) villi
- (C) stomata
- (D) nephrons

30. The decimal 3.45 rounded to one decimal place (the nearest tenth) is:

- (A) 4.5
- (B) 3.0
- (C) 3.4
- (D) 3.5

- 31.** The waste liquid made by the kidneys and stored before being passed out is:
- (A) urine
 - (B) plasma
 - (C) bile
 - (D) sweat
- 32.** A square photo frame is built from 36 cm of wooden beading all around. Each side of the frame is:
- (A) 9 cm
 - (B) 18 cm
 - (C) 36 cm
 - (D) 4 cm
- 33.** Insects such as a grasshopper take in air through small openings on the body called:
- (A) skin pores
 - (B) spiracles
 - (C) gills
 - (D) lungs
- 34.** The product 0.2×0.3 equals:
- (A) 6
 - (B) 0.5
 - (C) 0.6
 - (D) 0.06
- 35.** Removal of waste products such as urea from the body is called:
- (A) circulation
 - (B) respiration
 - (C) digestion
 - (D) excretion
- 36.** The width of a rectangle of area 48 cm^2 and length 8 cm is:
- (A) 6 cm
 - (B) 16 cm
 - (C) 40 cm
 - (D) 384 cm
- 37.** Fish are able to breathe in water because they take in dissolved oxygen using their:
- (A) gills
 - (B) lungs
 - (C) spiracles
 - (D) skin only

38. The value of $4.5 / 0.5$ is:

- (A) 2.25
- (B) 0.9
- (C) 90
- (D) 9

39. Water and minerals move up from the roots to the leaves of a plant through the:

- (A) cambium
- (B) stomata
- (C) phloem
- (D) xylem

40. A rectangle measuring 10 cm by 4 cm is reshaped into a square using the same length of boundary. The square's side becomes:

- (A) 28 cm
- (B) 7 cm
- (C) 14 cm
- (D) 10 cm

41. An earthworm has no lungs or gills; it exchanges gases through its:

- (A) lungs
- (B) spiracles
- (C) moist skin
- (D) gills

42. Three-fifths written as a decimal is:

- (A) 0.06
- (B) 0.35
- (C) 0.5
- (D) 0.6

43. The food made in the leaves is carried to all other parts of the plant through the:

- (A) xylem
- (B) bark
- (C) roots
- (D) phloem

44. A path of area 30 m^2 costs ₹20 per square metre to pave. The total cost is:

- (A) ₹600
- (B) ₹6000
- (C) ₹150
- (D) ₹50

45. During exhalation in humans, the air breathed out is richer in:
- (A) hydrogen
 - (B) oxygen
 - (C) nitrogen
 - (D) carbon dioxide
46. The improper fraction $\frac{7}{2}$ written as a mixed number is:
- (A) $3\frac{1}{3}$
 - (B) $2\frac{1}{3}$
 - (C) $3\frac{1}{2}$
 - (D) $7\frac{1}{2}$
47. The loss of water as vapour from the leaves of a plant is called:
- (A) germination
 - (B) transpiration
 - (C) respiration
 - (D) condensation
48. A wire of length 40 cm is bent into a square. The area of the square is:
- (A) 10 cm^2
 - (B) 40 cm^2
 - (C) 100 cm^2
 - (D) 160 cm^2
49. Plants carry out gas exchange in their leaves mainly through tiny pores called:
- (A) lenticels only
 - (B) stomata
 - (C) spiracles
 - (D) alveoli
50. The value of $\frac{2}{5}$ of 60 is:
- (A) 12
 - (B) 24
 - (C) 150
 - (D) 30
51. The pull created by transpiration helps the plant by:
- (A) making food without sunlight
 - (B) drawing water and minerals up from the roots
 - (C) killing harmful germs
 - (D) storing extra sugar
52. A leaf is shaped like a rectangle 4 cm by 3 cm. Its surface area on one side is:
- (A) 7 cm^2
 - (B) 12 cm^2
 - (C) 12 cm
 - (D) 14 cm^2

53. Both day and night, all living cells of a plant carry out:
- (A) respiration
 - (B) transpiration only
 - (C) germination
 - (D) photosynthesis
54. The decimal 0.125 written as a fraction in lowest terms is:
- (A) $\frac{1}{8}$
 - (B) $\frac{125}{10}$
 - (C) $\frac{1}{4}$
 - (D) $\frac{1}{5}$
55. In humans, blood is filtered to remove wastes by a pair of bean-shaped organs called the:
- (A) heart
 - (B) kidneys
 - (C) liver
 - (D) lungs
56. A rectangular field is 50 m long and 30 m wide. The length of fence to go around it once is:
- (A) 80 m
 - (B) 160 m
 - (C) 320 m
 - (D) 1500 m
57. The normal breathing rate of a resting healthy adult is about:
- (A) 100-110 times per minute
 - (B) 60-70 times per minute
 - (C) 2-4 times per minute
 - (D) 15-18 times per minute
58. Which decimal is the smallest: 0.3, 0.03, 0.33, 0.303?
- (A) 0.303
 - (B) 0.33
 - (C) 0.3
 - (D) 0.03
59. A doctor counts 72 heartbeats in one minute. In half a minute the heart beats about:
- (A) 18 times
 - (B) 36 times
 - (C) 70 times
 - (D) 144 times
60. The area of a square garden of side 0.5 m is:
- (A) 0.25 m^2
 - (B) 2 m^2
 - (C) 1 m^2
 - (D) 0.5 m^2

61. When you exercise hard, your breathing rate increases mainly to:
- (A) make more sweat
 - (B) warm up the lungs
 - (C) supply more oxygen and remove more carbon dioxide
 - (D) digest food faster
62. The sum $0.75 + 0.5$ equals:
- (A) 12.5
 - (B) 1.25
 - (C) 1.20
 - (D) 0.80
63. A heart beats 75 times a minute. The number of beats in 3 minutes is:
- (A) 150
 - (B) 225
 - (C) 250
 - (D) 78
64. A rectangular plant tray is 1.2 m by 0.5 m. Its area is:
- (A) 1.7 m^2
 - (B) 0.06 m^2
 - (C) 0.6 m^2
 - (D) 6 m^2
65. A boy breathes 18 times each minute at rest. The number of breaths in 5 minutes is:
- (A) 85
 - (B) 23
 - (C) 180
 - (D) 90
66. The reciprocal of $\frac{3}{7}$ is:
- (A) $\frac{7}{3}$
 - (B) $\frac{1}{3}$
 - (C) $-\frac{3}{7}$
 - (D) $\frac{3}{7}$
67. Of 80 heartbeats counted, $\frac{3}{4}$ felt strong and the rest faint. The number of strong beats is:
- (A) 20
 - (B) 75
 - (C) 40
 - (D) 60
68. Half of a rectangular leaf 6 cm by 4 cm is shaded. The shaded area is:
- (A) 6 cm^2
 - (B) 24 cm^2
 - (C) 12 cm^2
 - (D) 10 cm^2

69. A girl takes 16 breaths a minute. If each breath moves 0.5 litre of air, the air moved in one minute is:

- (A) 8 litres
- (B) 16.5 litres
- (C) 4 litres
- (D) 32 litres

70. A rope is 2.5 m long. If 0.75 m is cut off, the remaining length is:

- (A) 1.75 m
- (B) 1.25 m
- (C) 3.25 m
- (D) 2.25 m

71. A plant loses 0.6 litre of water by transpiration each day. In one week (7 days) it loses:

- (A) 42 litres
- (B) 0.42 litres
- (C) 4.2 litres
- (D) 6.7 litres

72. A square plot has an area of 81 m². The length of its side is:

- (A) 81 m
- (B) 9 m
- (C) 18 m
- (D) 40.5 m

73. A patient breathes 20 times a minute. The fraction of an hour taken by one minute of this breathing is:

- (A) $\frac{1}{100}$
- (B) $\frac{1}{24}$
- (C) $\frac{1}{60}$
- (D) $\frac{1}{20}$

74. The value of $1\frac{1}{2} + 2\frac{1}{4}$ is:

- (A) $3\frac{2}{6}$
- (B) $3\frac{3}{4}$
- (C) $3\frac{1}{4}$
- (D) 4

75. A patient's normal pulse is 72 and after a walk it is 90. The increase as a fraction of the normal pulse is:

- (A) $\frac{1}{4}$
- (B) $\frac{1}{5}$
- (C) $\frac{18}{90}$
- (D) $\frac{1}{3}$

76. A rectangular pond cover is $\frac{3}{4}$ m long and $\frac{1}{2}$ m wide. Its area is:

- (A) $\frac{3}{8} \text{ m}^2$
- (B) $\frac{3}{4} \text{ m}^2$
- (C) $\frac{1}{2} \text{ m}^2$
- (D) $\frac{5}{6} \text{ m}^2$

77. Out of 25 breaths, a child breathed only through the mouth for 5 of them. The fraction of mouth breaths is:

- (A) $\frac{1}{5}$
- (B) $\frac{1}{4}$
- (C) $\frac{5}{20}$
- (D) $\frac{5}{100}$

78. The decimal 0.6 written as a percentage is:

- (A) 600%
- (B) 6%
- (C) 0.6%
- (D) 60%

79. In 1 litre of blood, 0.45 litre is red cells. Written as a percentage, the red-cell part is:

- (A) 0.45%
- (B) 4.5%
- (C) 45%
- (D) 55%

80. A garden bed of area 24 m^2 has $\frac{1}{3}$ of it planted with roses. The rose area is:

- (A) 12 m^2
- (B) 21 m^2
- (C) 8 m^2
- (D) 3 m^2

81. A diver's air tank holds 12 litres and he uses $\frac{3}{4}$ of it. The volume of air used is:

- (A) 9 litres
- (B) 4 litres
- (C) 8 litres
- (D) 3 litres

82. The value of $0.4 + 0.45$ is:

- (A) $0.85 / 2$
- (B) 0.85
- (C) 0.49
- (D) 4.45

83. A tree's xylem carries 8 litres of water up daily and $\frac{3}{8}$ of it reaches the topmost leaves. The water reaching the top is:

- (A) 3 litres
- (B) 5 litres
- (C) 11 litres
- (D) 8 litres

84. Two identical square tiles of side 5 cm are placed side by side to form a rectangle. The rectangle's area is:

- (A) 100 cm^2
- (B) 25 cm^2
- (C) 20 cm^2
- (D) 50 cm^2

85. In limewater 0.4 of the bubbles came from carbon dioxide. Written as a percentage this is:

- (A) 400%
- (B) 0.4%
- (C) 40%
- (D) 4%

86. What fraction of 1 hour is 20 minutes?

- (A) $\frac{1}{20}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{20}{100}$

87. Blood plasma is about $\frac{9}{10}$ water. Written as a decimal, this fraction is:

- (A) 1.9
- (B) 9.0
- (C) 0.9
- (D) 0.09

88. A square field of side 20 m needs fencing on all sides. At ₹15 per metre, the fencing cost is:

- (A) ₹1380
- (B) ₹6000
- (C) ₹1200
- (D) ₹300

89. A resting man breathes 15 times a minute; while jogging he breathes 24 times. The increase as a fraction of the resting rate is:

- (A) $\frac{3}{5}$
- (B) $\frac{9}{24}$
- (C) $\frac{2}{5}$
- (D) $\frac{3}{8}$

90. The value of $\frac{5}{6} - \frac{1}{3}$ is:

- (A) $\frac{1}{2}$
- (B) $\frac{4}{6}$
- (C) $\frac{4}{3}$
- (D) $\frac{1}{3}$

91. A heart pumps 0.07 litre of blood per beat. In 100 beats it pumps:

- (A) 70 litres
- (B) 0.7 litres
- (C) 7 litres
- (D) 17 litres

92. The base of a triangle whose area is 24 cm^2 and height is 8 cm is:

- (A) 3 cm
- (B) 16 cm
- (C) 12 cm
- (D) 6 cm

93. A fish tank uses 2.5 litres of fresh water per hour for aeration. In 4 hours it uses:

- (A) 6.5 litres
- (B) 10 litres
- (C) 8 litres
- (D) 9 litres

94. Anu spent 0.35 of her money and saved the rest. The decimal part she saved is:

- (A) 0.65
- (B) 0.35
- (C) 0.75
- (D) 1.35

95. Of 1 litre of blood, red cells are 0.45 L and plasma 0.55 L. The plasma is more than the red cells by:

- (A) 0.2 litre
- (B) 0.9 litre
- (C) 1.0 litre
- (D) 0.1 litre

96. A leaf-shaped paper covers $\frac{3}{5}$ of a rectangular sheet of area 100 cm^2 . The leaf area is:

- (A) 60 cm^2
- (B) 40 cm^2
- (C) 35 cm^2
- (D) 20 cm^2

97. A runner's breaths-per-minute went from 16 at rest to 28 while sprinting. The total rise in breaths per minute is:

- (A) 28
- (B) 44
- (C) 12
- (D) 10

98. The value of $\frac{7}{10}$ written as a decimal is:

- (A) 0.7
- (B) 0.07
- (C) 0.17
- (D) 7.0

99. The watery, pale-yellow part of blood that carries cells and dissolved food is called:

- (A) urea
- (B) bile
- (C) plasma
- (D) haemoglobin

100. A rectangular vegetable patch is 9 m long and 4 m wide. How much greater is its area than its perimeter (as numbers)?

- (A) 13
- (B) 10
- (C) 62
- (D) 36